

The General Eberlein–Grothendieck/ Hemicompactness Equivalence

Candidate full solution for arXiv:2206.10684

13 August 2026

Status. Candidate full solution, likely valid; human review required. The theorem below removes the first-countability hypothesis from the result of Kąkol–Leiderman. Novelty is provisional pending specialist review.

1 Source problem

Kąkol and Leiderman study, for a Tychonoff space X , the locally convex space $C_k(X)$ of real-valued continuous functions with the compact-open topology. They write w for its weak topology and ask the following in Problem 2.6 of [1] (PDF p. 5):

Is $(C_k(X), w)$ Eberlein–Grothendieck if and only if X is hemicompact?

Here a space is Eberlein–Grothendieck if it embeds into $C_p(L)$ for some compact L , and X is *hemicompact* if it has a sequence of compact subsets cofinal among all compact subsets. The source proves the equivalence when X is first countable. Its exact statement and scope are shown below.

not σ -compact. Therefore, φ endowed with the weak topology is not Eberlein–Grothendieck, by Proposition 2.1.

For the brevity let us denote by EG_k the class of spaces X such that a lcs $(C_k(X), w)$ is Eberlein–Grothendieck. It is known that $C_k(X)$ is metrizable if and only if X is hemicompact (see for instance [15, Theorem 5.8.5]). Then, according to Corollary 2.3, $X \in EG_k$ in the case that X is hemicompact.

Problem 2.6. *Is the converse true, i.e. are the following properties equivalent: (1) $(C_k(X), w)$ is an Eberlein–Grothendieck lcs and (2) X is hemicompact?*

In this paper we show that the answer to Problem 2.6 is positive for all first-countable spaces X .

Figure 1: Problem 2.6 in [1], together with the source’s first-countable scope.

2 Proof intuition

The source reduces the Eberlein–Grothendieck condition to weak-star σ -compactness of the dual. Its dual elements are compactly supported Radon measures. The key new observation is that the

supports of a weak-star compact family cannot collectively escape along the level sets of a continuous function: if they did, support-localized test functions in disjoint high bands could be weighted and summed to form one continuous function on which the compact family has unbounded evaluations.

Thus every compact piece of the dual has functionally bounded total support. The Eberlein–Grothendieck hypothesis already forces X itself to be σ -compact. Such a Tychonoff space is a paracompact μ -space, so a functionally bounded subset has compact closure. An increasing compact cover of the dual therefore gives increasing compact support envelopes H_n in X .

The remaining issue is cofinality, not coverage. If a compact K escaped every H_n , choose $x_n \in K \setminus H_n$ and form the positive measure $\sum 2^{-n} \delta_{x_n}$. It belongs to some fixed dual-cover piece, forcing all its atoms into that piece’s support envelope. Monotonicity says that all sufficiently late atoms lie outside the same envelope, a contradiction.

3 Preliminaries

We use three results recorded in the source [1].

1. For a locally convex space E , (E, w) is Eberlein–Grothendieck exactly when (E^*, w^*) is σ -compact (Proposition 2.1).
2. Every $\phi \in C_k(X)^*$ is represented by a unique Radon measure μ on the Hewitt realcompactification νX whose support is a compact subset of X (Theorem 2.7, quoted from [3]). We identify $C_k(X)^*$ with this measure space $\mathcal{M}_c(X)$.
3. The Dirac map $x \mapsto \delta_x$ is a homeomorphic closed embedding of X into the weak-star dual

$$(\mathcal{M}_c(X), \sigma(\mathcal{M}_c(X), C(X)))$$

(Proposition 2.8).

We also use the support-localization clause of the measure representation: if $U \subset \nu X$ is open and meets $\text{supp } \mu$, there is $g \in C(X)$ that vanishes on $\nu X \setminus U$ and satisfies $\mu(g) = 1$.

Recall that $B \subset X$ is *functionally bounded* if every member of $C(X)$ is bounded on B .

Lemma 1 (Paracompact μ -space fact). *If X is paracompact Hausdorff, then the closure of every functionally bounded subset of X is compact.*

Proof. Let B be functionally bounded and put $F = \overline{B}^X$. If F were noncompact, then the paracompact Hausdorff space F would not be pseudocompact: a paracompact pseudocompact Hausdorff space is compact. Consequently some $h \in C(F)$ would be unbounded. The ambient space X is normal, and the real-valued Tietze extension theorem extends h to a member of $C(X)$. Since B is dense in F , continuity makes h unbounded already on B , contrary to functional boundedness. Hence F is compact. $\square$

4 The support-envelope lemma

Lemma 2. *Let X be Tychonoff and let A be a weak-star compact subset of $C_k(X)^*$. Then*

$$S_A := \bigcup_{\mu \in A} \text{supp } \mu$$

is functionally bounded in X .

Proof. Suppose not. Choose $u \in C(X)$ such that $|u|$ is unbounded on S_A , and let $q = |\tilde{u}| \in C(\nu X)$, where $\tilde{u}$ is the extension of u to the Hewitt realcompactification.

We inductively choose measures $\mu_n \in A$, points $x_n \in \text{supp } \mu_n$, and pairwise disjoint open bands

$$U_n = \{y \in \nu X : a_n < q(y) < b_n\}.$$

At stage n , all earlier supports are compact, so q is bounded on their finite union. Since q is unbounded on S_A , choose x_n so high that the interval (a_n, b_n) containing $q(x_n)$ has its lower endpoint above the previous bands, above the maximum of q on every earlier support, and above n . Choose $\mu_n \in A$ with $x_n \in \text{supp } \mu_n$. At the next stage also place the new lower endpoint above $\max_{\text{supp } \mu_n} q$. It follows that

$$U_j \cap \text{supp } \mu_i = \emptyset \quad (j > i). \quad (1)$$

The lower endpoints tend to infinity, so (U_n) is locally finite.

By support localization, for every n there is $g_n \in C(X)$ vanishing off U_n and satisfying $\mu_n(g_n) = 1$. Choose real numbers c_n recursively. After $c_1, \dots, c_{n-1}$ are fixed, take

$$c_n = n - \sum_{j < n} c_j \mu_n(g_j).$$

Because the bands are locally finite,

$$g := \sum_{n=1}^{\infty} c_n g_n$$

is a well-defined continuous function on X . By (1), every term g_j with $j > n$ vanishes on $\text{supp } \mu_n$. Hence

$$\mu_n(g) = \sum_{j \leq n} c_j \mu_n(g_j) = n.$$

On the other hand, evaluation at g is a continuous real-valued function on the weak-star compact space A , so it must be bounded. This contradiction proves the lemma. $\square$

Remark 3. *Lemma 2 deliberately concludes only functional boundedness. Without additional hypotheses on X , compactness of $\overline{S_A}^X$ can fail; this is why the σ -compactness forced by the Eberlein–Grothendieck assumption is used separately below.*

5 Full solution

Theorem 4. *For every Tychonoff space X , the following are equivalent:*

1. $(C_k(X), w)$ is Eberlein–Grothendieck;
2. X is hemicompact.

Proof. Assume first that $(C_k(X), w)$ is Eberlein–Grothendieck. By the source’s Proposition 2.1, the weak-star dual $\mathcal{M}_c(X)$ is σ -compact. Choose compact sets whose union is $\mathcal{M}_c(X)$ and replace them by their finite unions; we obtain an increasing compact cover

$$A_1 \subset A_2 \subset \cdots, \quad \mathcal{M}_c(X) = \bigcup_{n=1}^{\infty} A_n.$$

The closed Dirac embedding of X into $\mathcal{M}_c(X)$ shows that X is σ -compact. A Tychonoff σ -compact space is Lindelöf and hence paracompact. For every n , define

$$H_n = \overline{\bigcup_{\mu \in A_n} \text{supp } \mu}^X.$$

Lemma 2 and Lemma 1 show that each H_n is compact. Moreover (H_n) is increasing.

We claim that (H_n) is cofinal among the compact subsets of X . If not, there is a compact $K \subset X$ with $K \not\subseteq H_n$ for every n . Choose

$$x_n \in K \setminus H_n$$

and consider the positive atomic measure

$$\mu = \sum_{n=1}^{\infty} 2^{-n} \delta_{x_n}.$$

This is a Radon probability supported in K . For every $f \in C(X)$,

$$|\mu(f)| \leq \sup_{x \in K} |f(x)|,$$

so μ is a continuous linear functional on $C_k(X)$, that is, $\mu \in \mathcal{M}_c(X)$. Choose N with $\mu \in A_N$. By definition of H_N ,

$$\text{supp } \mu \subset H_N.$$

Every x_n is a positive atom of μ , so $x_n \in \text{supp } \mu$. But if $n \geq N$, then $H_N \subset H_n$ and $x_n \notin H_n$, hence $x_n \notin H_N$. This is a contradiction. Therefore (H_n) is cofinal and X is hemicompact.

Conversely, if X is hemicompact, then $C_k(X)$ is metrizable [2, Theorem 5.8.5]. Every metrizable locally convex space endowed with its weak topology is Eberlein–Grothendieck [1, Corollary 2.3]. Hence $(C_k(X), w)$ is Eberlein–Grothendieck. $\square$

6 Verification, scope, and novelty

The proof uses no first-countability, sequentiality, separability, or local compactness. The potentially delicate points are as follows.

- The support bands in Lemma 2 are locally finite because their lower levels tend to infinity; arbitrary coefficients therefore do not endanger continuity of their sum.
- The induction gives a triangular vanishing relation, so future summands vanish on each earlier compact support. There is no interchange of an uncontrolled infinite sum and a signed measure.
- The diagonal measure is positive. Thus every chosen point is an atom in its support; no compact set is assumed to carry a measure of full support.

- Functional boundedness is upgraded to compact closure only after the EG hypothesis forces X to be σ -compact and paracompact.

The run's lightweight indexes contained no earlier result for arXiv:2206.10684. Bounded exact-title, exact-problem, notation, author, hemicompactness, and weak-star-support searches through 13 August 2026 found the source, a conference presentation repeating the first-countable result, and later citations, but no explicit resolution of Problem 2.6. The argument should therefore be treated as plausibly new, not as definitively novel, until an expert literature review is completed.

7 Human-review recommendation

The candidate should be reviewed as a full solution. The highest-value checks are: (i) the support-localization theorem in the precise form used in Lemma 2; (ii) the paracompact μ -space fact; and (iii) the support inclusion and monotonicity in the final atomic diagonal. No computational verification is relevant.

References

- [1] J. Kąkol and A. Leiderman, *When is a locally convex space Eberlein–Grothendieck?*, arXiv:2206.10684 (2022).
- [2] L. Narici and E. Beckenstein, *Topological Vector Spaces*, 2nd ed., Chapman and Hall/CRC, 2010.
- [3] J. Schmets, *Espaces de fonctions continues*, Lecture Notes in Mathematics 519, Springer, 1976.